

Autonomous Shuttle and BRT Integration: An Agent-Based Case Study of Paris's Southern Suburbs

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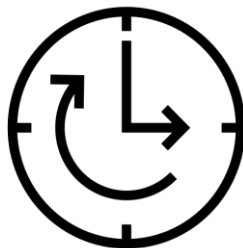
MATSim User Meeting 2026
2026/09/28

Context

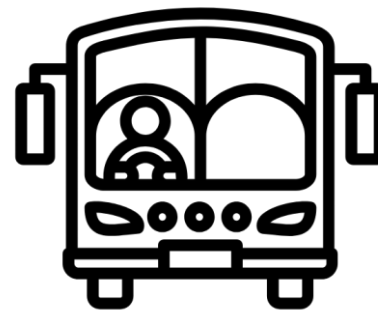
Shared autonomous vehicles (SAVs) as a solution for mobility challenges



Reduce labor costs



Extend service hours



Mitigate human driving errors

(Carrese et al., 2023)

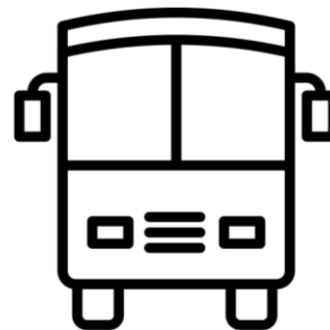
Context



SAVs

Rivals?

and



Public transport



Modal shift

(Carrese et al., 2023)

Icons by Hanbai and ainul muttaqin from Noun Project

Context



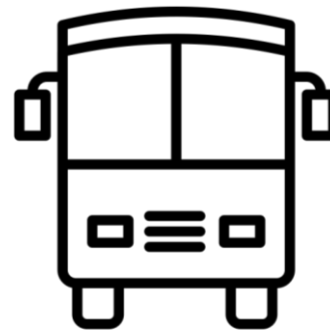
SAVs

Flexible

Low-demand areas

Off-peak hours

complements



Public transport

High capacity

Dense corridors

(Carrese et al., 2023)

Research question

- How can SAVs complement, rather than compete with, existing transit systems?

More specifically, in a suburban context

- How can autonomous shuttles be used most effectively on suburban highways?
 - High demand in peak hours and very low demand in off-peak hours

Case study

MOBAUTO2 project

MOBilité AUTOMatisée sur AUTOroute connectée



12-seat autonomous shuttles



Car Express 91.03 Bus Rapid Transit (BRT)
86 seats

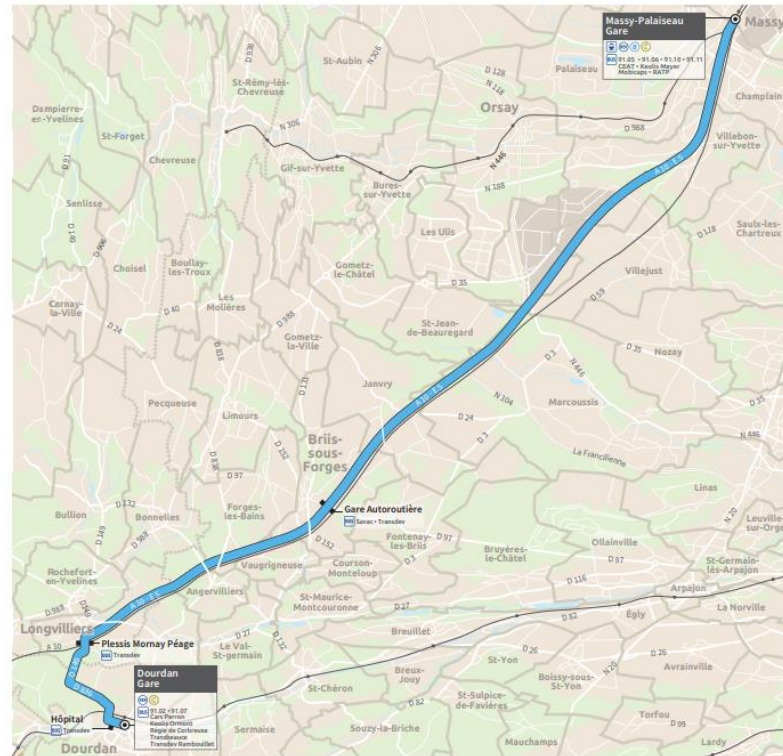
Case study

BRT:

Peak hours: capacity constraints

Off-peak hours: limited service

BUS Express 91.03 DOURDAN ↔ MASSY-PALAISEAU



Icon by supalerk laipawat from Noun Project

Case study

BRT:

Peak hours: capacity constraints

Off-peak hours: limited service

Could the autonomous shuttles enhance the area's public transit service by alleviating the load on the BRT line?



Icon by supalerk laipawat from Noun Project

Case study

Scenario design

Business
as usual
(BAU)

The BRT line
operates without
any modification

Complementary
service
(CS)

The AS supplements the
BRT line. Two
autonomous shuttle
(AS) departures are
added between each
BRT departure

Partial
substitution
(CS)

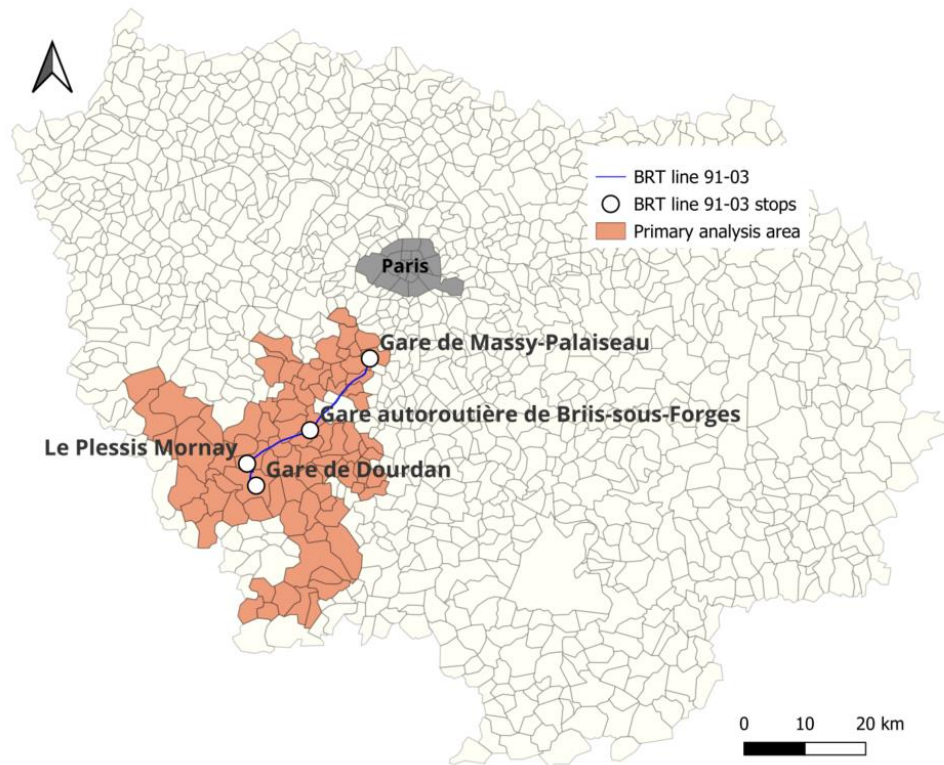
Autonomous shuttles
replace low-frequency
BRT trips during off-peak
periods while maintaining
BRT service during peaks
to handle high demand

Case study

Primary analysis area:

BAU + Extreme simulation scenario
(1-minute bus headway)

Residential and workplace
municipalities with +40 BRT users in
Essonne and Yvelines departments
south of the Hauts-de-Seine
department, with few exceptions



Methodology

Simulation environment

- MATSIM + Discrete Mode Choice (DMC) module (Hörl & Balac, 2021)
1 run, 100 iterations
- Synthetic population: 10% of Ile-de-France region
Sociodemographic attributes and a daily activity patterns
eqasim pipeline (Hörl & Balac, 2021)
- Autonomous shuttle service implemented in MATSim as a scheduled public transport service integrated into the existing transit network

A stated-preference survey concluded that autonomous shuttles are perceived as conventional modes of public transportation rather than as a distinct category.

Results

Overall behavior

Mode shares by number of trips

	All Île-de-France					Primary analysis area				
	Car	Car pass.	PT	Walk	Bike	Car	Car pass.	PT	Walk	Bike
Travel survey (EGT 2010)	30,7%	8,1%	20,6%	39,0%	1,59%	49,0%	13,0%	12,0%	23,0%	0,9%
BAU scenario	31,8%	7,9%	17,8%	41,0%	1,54%	44,0%	13,3%	8,9%	31,4%	2,3%

Results

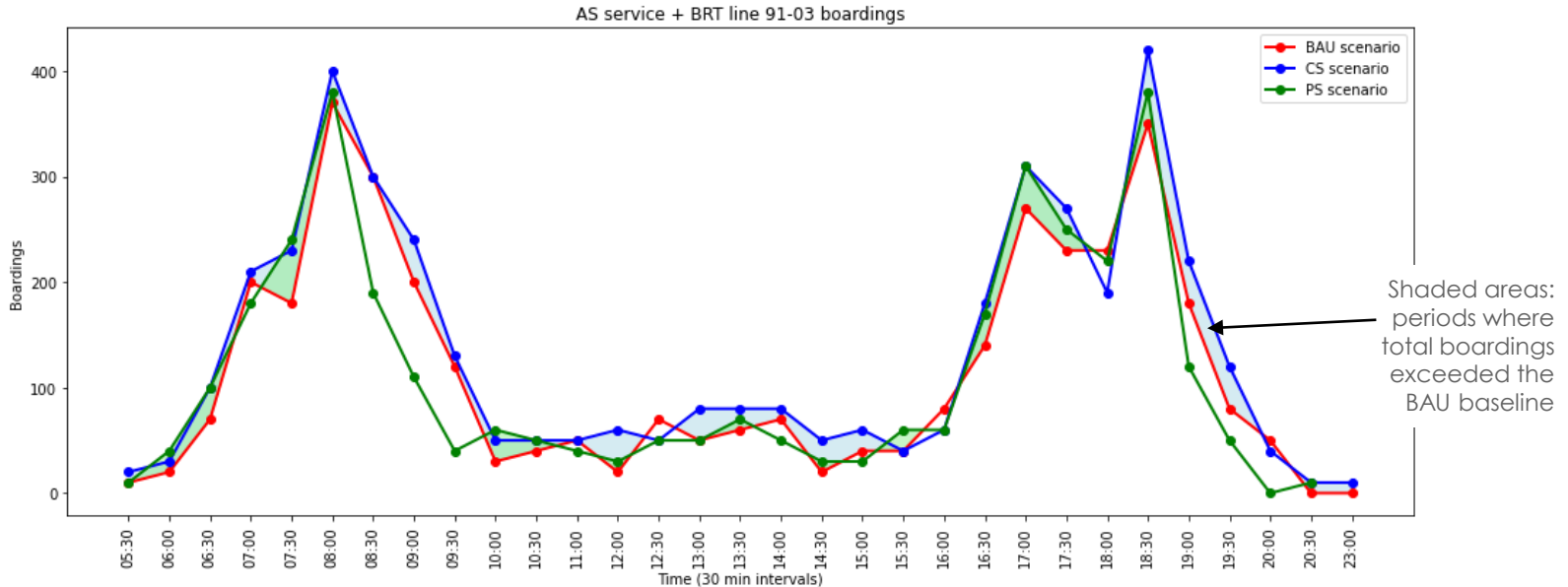
Corridor usage

	Day boardings		
	BRT line	AS service	Total
IDFM data (mean*)	3202	---	3202
BAU scenario	3570	---	3570
CS scenario	2840	1300	4140
PS scenario	1520	1860	3380

*Tuesdays/Thursdays, March–June 2024

Results

Corridor usage

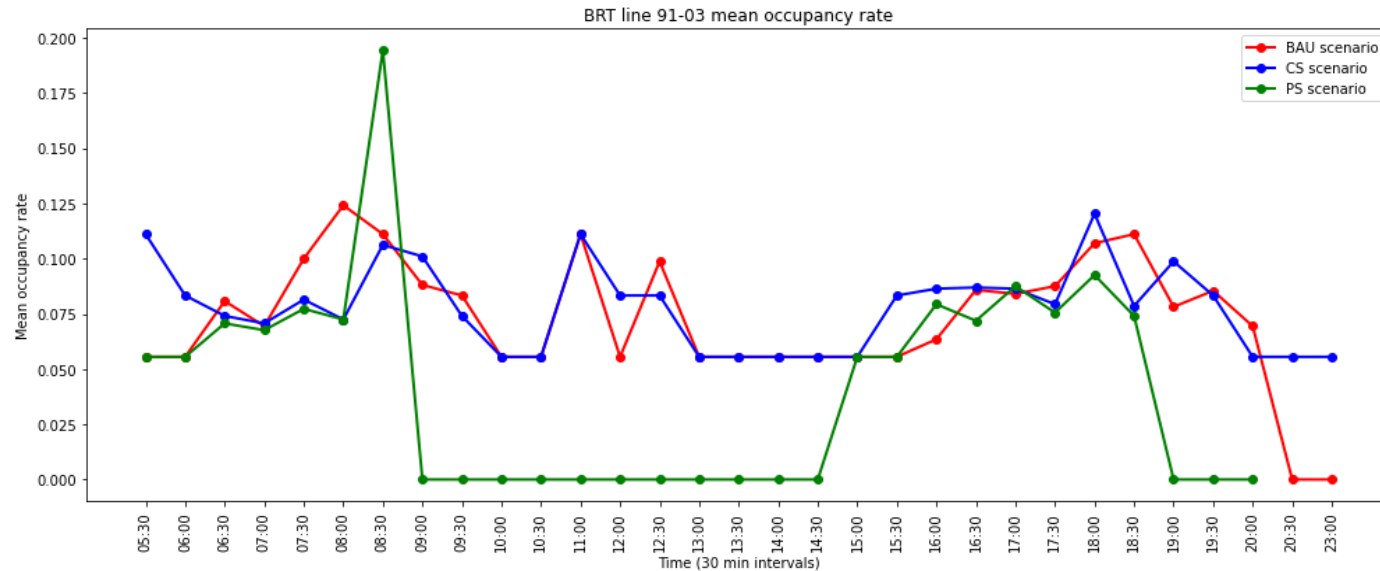


Number of boardings comparison between scenarios

Results

Shuttle & BRT usage

Capacities
(10% synthetic pop):
BRT: 90 seats
AS: 20 seats



Mean occupancy rate comparison between scenarios
BRT

Results

Shuttle & BRT usage

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Mean occupancy rate comparison between scenarios
AS service

Conclusions

Key finding

The current configurations in both CS and PS scenarios result in only limited ridership shifts.

Limitations and directions for future research

- Size of synthetic population: we used a very small synthetic population (10%), limiting the scope of the proposed scenarios
→ change it to 25%
- AS as demand-responsive transport using the DRT module with a stop-based operational scheme

Thank you for your attention

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References

Carrese, F., Sportiello, S., Zhaksylykov, T., Colombaroni, C., Carrese, S., Papaveri, M., & Patella, S. M. (2023). The integration of shared autonomous vehicles in public transportation services: A systematic review. *Sustainability*, 15(17), 13023.

Hörl, S., & Balac, M. (2021). Synthetic population and travel demand for Paris and Île-de-France based on open and publicly available data. *Transportation Research Part C: Emerging Technologies*, 130, 103291.